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Studierichting: Informatica
Oefeningenreeks 3H nr. 10.2

$$r = \sin\frac{\theta}{2}$$

$$\text{dom} = \mathbb{R}$$

$$p = 4\pi$$

$$\rightarrow \text{BD} = [0, 4\pi]$$

$$r = 0 \Leftrightarrow \sin\frac{\theta}{2} = 0 \Leftrightarrow \frac{\theta}{2} = k\pi \Leftrightarrow \theta = 2k\pi, \forall k \in \mathbb{Z}$$

$$\rightarrow \{0, 2\pi, 4\pi\}$$

$$r' = 0 \Leftrightarrow \cos\frac{\theta}{2} = 0 \Leftrightarrow \frac{\theta}{2} = \frac{\pi}{2} + k\pi \Leftrightarrow \theta = \pi + 2k\pi, \forall k \in \mathbb{Z}$$

$$\rightarrow \{\pi, 3\pi\}$$

θ	0		π		2π		3π		4π
r	0	$+$	$+$	$+$	0	$-$	$-$	$-$	0
r'	$+$	$+$	0	$-$	$-$	$-$	0	$+$	$+$
r	$\nearrow$	$\nearrow$	M	$\searrow$	$\searrow$	$\searrow$	m	$\nearrow$	$\nearrow$
			1				-1		

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References